

Principia, besides being a lasting monument of Newton's life, is also to-day the classic of our mathematical writings, and will be so for some time to come. . . . The value of the present work is also enhanced by the fact that, besides containing a few as yet unpublished letters, there are collected in its pages quotations from all documents, thus forming a complete summary of everything that is known on the subject. . . . The author is so well known a writer on anything connected with the history of mathematics, that we need make no mention of the thoroughness of the essay, while it would be superfluous for us to add that from beginning to end it is pleasantly written and delightful to read. Those well acquainted with the *Principia* will find much that will interest them, while those not so fully enlightened will learn much by reading through the account of the origin and history of Newton's greatest work.—*Nature*.

An Essay on Newton's Principia will suggest to many something solely mathematical, and therefore wholly uninteresting. No inference could be more erroneous. The book certainly deals largely in scientific technicalities which will interest experts only; but it also contains much historical information which might attract many who, from laziness or inability, would be very willing to take all its mathematics for granted. Mr. Ball carefully examines the evidence bearing on the development of Newton's great discovery, and supplies the reader with abundant quotations from contemporary authorities. Not the least interesting portion of the book is the appendix, or rather appendices, containing copies of the original documents (mostly letters) to which Mr. Ball refers in his historical criticisms. Several of these bear upon the irritating and unfounded claims of Hooke.—*The Athenaeum*.

La savante monographie de M. Ball est rh(=—e('digh(=—e('e avec beaucoup de soin, et h(=—a(=— plusieurs h(=—e('gards elle peut servir de modh(=—e)'le pour des h(=—e('crits de la même nature.—*Bibliotheca Mathematica*.

Newton's *Principia* has world-wide fame as a classic of mathematical science. But those who know thoroughly the contents and the history of the book are a select company. It was at one time the purpose of Mr. Ball to prepare a new critical edition of the work, accompanied by a prefatory history and notes, and by an analytical commentary. Mathematicians will regret to hear that there is no prospect in the immediate future of seeing this important book carried to completion by so competent a hand. They will at the same time welcome Mr. Ball's *Essay on the Principia* for the elucidations which it gives of the process by which Newton's great work originated and took form, and also as an earnest of the completed plan.—*The Scotsman*.

In this essay Mr. Ball presents us with an account highly interesting to mathematicians and natural philosophers of the origin and history of that remarkable product of a great genius *Philosophiae Naturalis Principia Mathematica*, 'The Mathematical Principles of Natural Philosophy,' better known by the short term *Principia*. . . . Mr. Ball's essay is one of extreme interest to students of physical science, and it is sure to be widely read and greatly appreciated.—*The Glasgow Herald*.

To his well-known and scholarly treatises on the *History of Mathematics* Mr. W.W. Rouse Ball has added *An Essay on Newton's Principia*. Newton's *Prin-*

cipia, as Mr. Ball justly observes, is the classic of English mathematical writings; and this sound, luminous, and laborious essay ought to be the classical account of the *Principia*. The essay is the outcome of a critical edition of Newton's great work, which Mr. Ball tells us that he once contemplated. It is much to be hoped that he will carry out his intention, for no English mathematician is likely to do the work better or in a more reverent spirit. . . . It is unnecessary to say that Mr. Ball has a complete knowledge of his subject. He writes with an ease and clearness that are rare.—*The Scottish Leader*.

Le volume de M. Rouse Ball renferme tout ce que l'on peut dh(=—e('sirer savoir sur l'histoire des *Principes*; c'est d'ailleurs l'œuvre d'un esprit clair, judicieux, et mh(=—e('thodique.—*Bulletin des Sciences Mathh(=—e('matiques*.

Mr. Ball has put into small space a very great deal of interesting matter, and his book ought to meet with a wide circulation among lovers of Newton and the *Principia*.—*The Academy*.

Admirers of Mr. W.W. Rouse Ball's *Short Account of the History of Mathematics* will be glad to receive a detailed study of the history of the *Principia* from the same hand. This book, like its predecessors, gives a very lucid account of its subject. We find in it an account of Newton's investigations in his earlier years, which are to some extent collected in the tract *de Motu* (the germ of the *Principia*) the text of which Mr. Rouse Ball gives us in full. In a later chapter there is a full analysis of the *Principia* itself, and after that an account of the preparation of the second and third editions. Probably the part of the book which will be found most interesting by the general reader is the account of the correspondence of Newton with Hooke, and with Halley, about the contents or the publication of the *Principia*. This correspondence is given in full, so far as it is recoverable. Hooke does not appear to advantage in it. He accuses Newton of stealing his ideas. His vain and envious disposition made his own merits appear great in his eyes, and be-dwarfed the work of others, so that he seems to have believed that Newton's great performance was a mere expanding and editing of the ideas of Mr. Hooke—ideas which were meritorious, but after all mere guesses at truth. This, at all events, is the most charitable view we can take of his conduct. Halley, on the contrary, appears as a man to whom we ought to feel most grateful. It almost seems as though Newton's physical insight and extraordinary mathematical powers might have been largely wasted, as was Pascal's rare genius, if it had not been for Halley's single-hearted and self-forgetful efforts to get from his friend's genius all he could for the enlightenment of men. It was probably at his suggestion that the writing of the *Principia* was undertaken. When the work was presented to the Royal Society, they undertook its publication, but, being without the necessary funds, the expense fell upon Halley. When Newton, stung by Hooke's accusations, wished to withdraw a part of the work, Halley's tact was required to avert the catastrophe. All the drudgery, worry, and expense fell to his share, and was accepted with the most generous good nature. It will be seen that both the technical student and the general reader may find much to interest him in Mr. Rouse Ball's book.—*The Manchester Guardian*.

Une histoire trh(=—e('s bien faite de la genh(=—e('se du livre immortel de Newton. . . . Le livre de M. Ball est une monographie prh(=—e('cieuse sur un point

important de l'histoire des mathématiques. Il contribuera à accroître, si c'est possible, la gloire de Newton, en rendant h(=a(= beaucoup de lecteurs, avec quelle merveilleuse rapidité l'illustre gh(=e('omh(=e)tre anglais a h(=e('levh(=e(' h(=a(= la science ce monument immortel, les *Principia*.—*Mathesis*.

NOTES ON THE HISTORY OF TRINITY COLLEGE, CAMBRIDGE

BY W.W. ROUSE BALL.

[*Pp.* xiv + 183. *Price* 2s. 6d. *net.*]

MACMILLAN AND CO. LTD., LONDON AND NEW YORK.

This booklet gives a popular account of the History of Trinity College, Cambridge, and so far as the author knows, it is as yet (1905) the only work published on the subject. It was written mainly for the use of his pupils, and contains such information and gossip about the College and life there in past times as he believed would be interesting to most undergraduates and members of the House.

This modest and unpretending little volume seems to us to do more for its subject than many of the more formal volumes . . . treating of the separate colleges of the English universities. . . . In nine short, extremely readable, and truly informing chapters it gives the reader a very vivid account at once of the origin and development of the University of Cambridge, of the rise and gradual supremacy of the colleges, of King's Hall as founded by Edward II, of the suppression of King's Hall by Henry VIII on December 17, 1546, the foundation of Trinity College by royal charter on December 19, and the subsequent fortunes of the premier college of Cambridge. The subject is in a way treated under the successive heads of the college, but this is quite subordinate to the handling and characterisation of the subject under four great periods—namely, that during the Middle Ages, that during the Renaissance, that under the Elizabethan statutes, and that during the last half-century. The colleges arose from the determination of the University to prevent students who were very young from seeking lodging, whether under the wing of one or other of the religious orders—a circumstance which shows this University to have been an essentially lay corporation. Early in the sixteenth century the college had absorbed all the members of the University, and henceforth the University was little more than the degree-granting body to students who lived and moved and had their educational being under the colleges. . . . The University finally took the form of an aggregate of separate and independent corporations, with a federal constitution analogous in a rough sort of way to that of the United States of America,

and different from similar corporations at Paris by the fact that these latter were always subject to University supervision. . . . There is a good account of the effort now going on to re-assert the University at the expense of the colleges. No one who begins Mr. Ball's book will lay it down till he has read it from beginning to end.—*The Glasgow Herald*.

It is a sign of the times, and a very satisfactory one, when . . . a tutor . . . takes the trouble to make the history of his college known to his pupils. Considering the lack of good books about the Universities, we may thank Mr. Ball that he has been good enough to print for a larger circle. Though he modestly calls his book only "Notes," yet it is eminently readable, and there is plenty of information, as well as abundance of good stories, in its pages.—*The Oxford Magazine*.

Mr. Ball has put not only the pupils for whom he compiled these notes, but the large world of Trinity men, under a great obligation by this compendious but lucid and interesting history of the society to whose service he is devoted. The value of his contribution to our knowledge is increased by the extreme simplicity with which he tells his story, and the very suggestive details which, without much comment, he has selected, with admirable discernment, out of the wealth of materials at his disposal. His initial account of the development of the University is brief but extremely clear, presenting us with facts rather than theories, but establishing, with much distinctness, the essential difference between the hostels, out of which the more modern colleges grew, and that monastic life which poorer students were often tempted to join.—*The Guardian*.

An interesting and valuable book. . . . It is described by its author as "little more than an orderly transcript" of what, as a Fellow and Tutor of the College, he has been accustomed to tell his pupils. But while it does not pretend either to the form or to the exhaustiveness of a set history, it is scholarly enough to rank as an authority, and far more interesting and readable than most academic histories are. It gives an instructive sketch of the development of the University and of the particular history of Trinity, noting its rise and policy in the earlier centuries of its existence, until, under the misrule of Bentley, it came into a state of disorder which nearly resulted in its dissolution. The subsequent rise of the College and its position in what Mr. Ball calls the Victorian renaissance, are drawn in lines no less suggestive; and the book, as a whole, cannot fail to be welcome to every one who is closely interested in the progress of the College.—*The Scotsman*.

Mr. Ball has succeeded very well in giving in this little volume just what an intelligent undergraduate ought and probably often does desire to know about the buildings and the history of his College. . . . The debt of the "royal and religious foundation" to Henry VIII is explained with fulness, and there is much interesting matter as to the manner of life and the expenses of students in the sixteenth century.—*The Manchester Guardian*.

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